

5.5 Substitution

- Calculating definite integrals using substitution. **HW 1-3, 6-7**
- This concept may be partially tested in problems from 6.1-6.3 and 6.5.
- Problems requiring substitution will not be labeled “use substitution”. Recognizing when to employ this technique is part of the problem.

6.1: Applications of Integration

- Given a derivative function, find the displacement or distance traveled over a given interval. **HW 1-2, 6, 9-10**
- Given a derivative function and initial position, find the displacement/position function. **HW 3-5, 8**

6.2: Area bounded by curves

- Find the area between curves. **HW 1-10**
- This may require finding the intersection points and using a sum of two or more integrals.

6.3: Volume of solids

- General slicing method. **HW 1-2**
 - Understand the described shape.
 - Place the object in the Cartesian plane such that the cross sections are lines perpendicular to the x -axis (or y -axis), and the height of these lines is determined by a function.
 - Determine an area function for the cross sections dependent on x (or y).
 - Write a definite integral using this area function.
- Disk/Washer method. **HW 2-8**
- Applying the FTC, and understanding the inverse relationship between differentiation and integration. **HW 1-5, 17-18**

6.5 Length of a curve

- Use the formula $L = \int_a^b \sqrt{1 + f'(x)^2} dx$ to find the length of $f(x)$ from $[a, b]$. **HW 1-7**

Some general notes.

- You will be given the necessary formulas from section 4.8.
- So that all the concepts can be covered in a 50 minute exam, problems will generally be computationally easier than the homework OR you will only be asked to find the definite integral giving the area/volume.
- The quiz and homework are good reviews. You can review past due HW assignments in MyMathLab (MML) by going to “Gradebook” and clicking on “Review”. Here you can practice more problems by clicking on “Similar Exercise”.